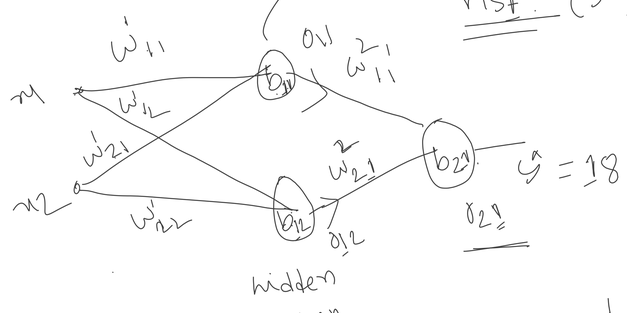
Backpropagation

Regression

Study hours	Sleep hours	H.S.C marks
2	3	3
5	6	4
7	8	5

Activation: linear

MSE



MSE: $(3-18)^2 = 225$

$= 9$

$o_{21} = w^2_{11} \cdot o_{11} + w^2_{21} \cdot o_{12} + b_{21}$

$o_{11} = w^1_{11} \cdot x_1 + w^1_{21} \cdot x_2 + b_{11}$

$o_{12} = w^1_{12} \cdot x_1 + w^1_{22} \cdot x_2 + b_{12}$

Steps:

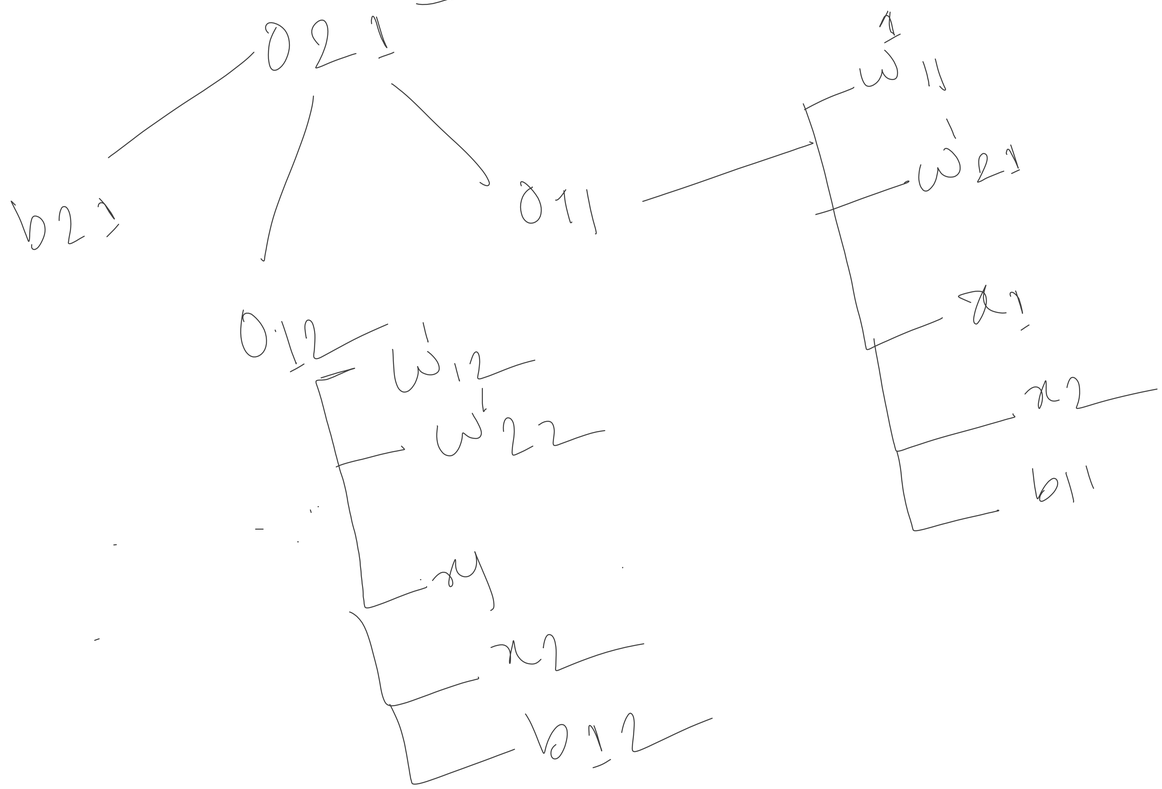
1) Initialization: w & b : $w = 1$, $b = 0$
2) predict → forward propagation
3) loss fn → apply
4) update w & b → using gradient descent

Dependency:

w^2_{11}

w^2_{21}

depe



formula:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

$$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}$$

$$w^2_{11} \rightarrow w^2_{11\,new} = w^2_{11\,old} - \eta \frac{\partial L}{\partial w^2_{11}}$$

$$w^2_{21} \rightarrow w^2_{21\,new} = w^2_{21\,old} - \eta \frac{\partial L}{\partial w^2_{21}}$$

$$b_{21\,new} = b_{21\,old} - \eta \frac{\partial L}{\partial}$$

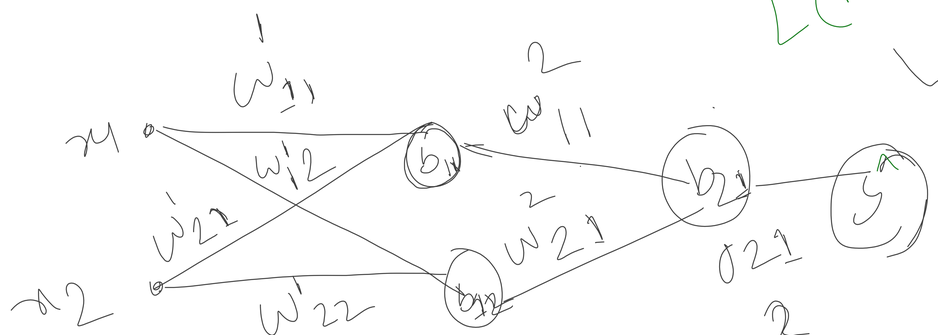
new = old

$b_{21} \rightarrow b_{21}^{new} = \ldots \quad \frac{\partial L}{\partial b_{21}}$

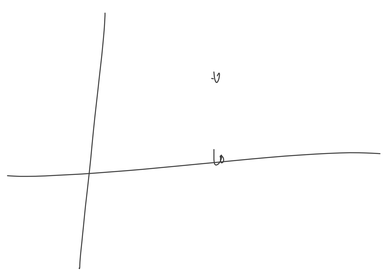
Let's calculate partial derivative:

$$\frac{\partial L}{\partial w^2_{11}} \quad \frac{\partial L}{\partial w^2_{21}} \quad \frac{\partial L}{\partial b_{21}} \quad \Big| \quad \frac{\partial L}{\partial w^1_{11}} \quad \frac{\partial L}{\partial w^1_{21}} \quad \frac{\partial L}{\partial b_{11}} \quad \Big| \quad \frac{\partial L}{\partial w^1_{12}} \quad \frac{\partial L}{\partial w^1_{22}} \quad \frac{\partial L}{\partial b_{12}}$$

$L(\min) \rightarrow \hat{y} \rightarrow o_{21} \rightarrow o_{11} \rightarrow w^1_{11}$



$o_{21} = w^2_{11} \cdot o_{11} + w^2_{21} \cdot o_{12} + b_{21}$



$\# \quad \frac{\partial L}{\partial w^2_{11}} = -2(y - \hat{y}) \cdot o_{11}$

∂w_{11}^{2}

$\hat{y}$

$L(\min)$

$A \to B \to C \to D \to E$

direct

indirect

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^{2}}$$

$$= (y-\hat{y})^2 \to -2(y-\hat{y})$$

$$\frac{\partial \hat{y}}{\partial w_{11}^{2}} = \frac{\partial}{\partial w_{11}^{2}}\left(w_{11}^{2} \cdot o_{11} + w_{21}^{2} \cdot o_{12} + b_{21}\right) = o_{11}$$

$$\# \quad \frac{\partial L}{\partial w_{21}^{2}} = -2(y-\hat{y}) \times o_{12}$$

$\hat{y} \to w_{21}^{2}$

$L(\min)$

$\partial L \quad \partial \hat{y}$

$$= \frac{\partial L}{\partial \hat{y}} \wedge \frac{\partial \hat{y}}{\partial w^2_{21}}$$

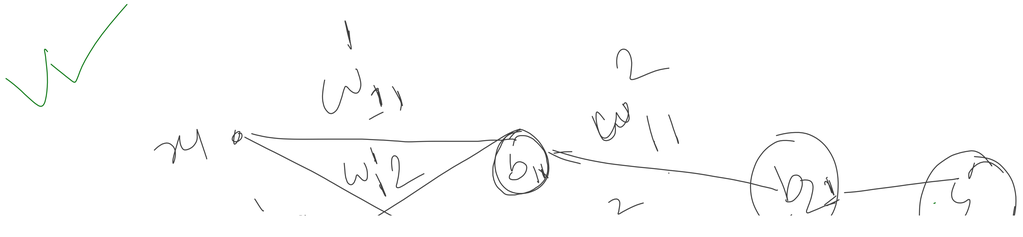
$$= -2(y-\hat{y}) \times O_{12}$$

$$\# \quad \frac{\partial L}{\partial b_{21}} = -2(y-\hat{y})$$

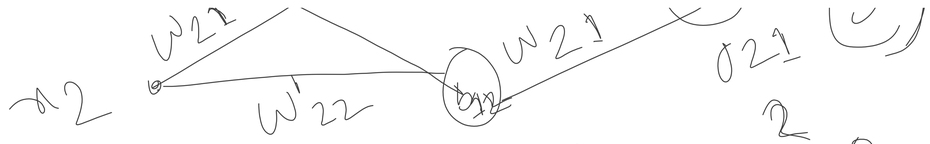
$$L(\min) \rightarrow \hat{y} \rightarrow b_{21}$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{21}}$$

$$= -2(y-\hat{y})$$



$$L(\min) \rightarrow \hat{y} \rightarrow O_{11} \rightarrow w^1_{11}$$



$$o_{21} = w^2_{11} \cdot o_{11} + w^2_{21} \cdot o_{12} + b_{21}$$

$$\# \frac{\partial L}{\partial w^1_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w^1_{11}}$$

$$\# \frac{\partial L}{\partial w^1_{21}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{11}} \frac{\partial o_{11}}{\partial w^1_{21}}$$

$$\# \frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{11}} \frac{\partial o_{11}}{\partial b_{11}}$$

$$\frac{\partial \hat{y}}{\partial o_{11}} = w^2_{11} \quad \checkmark$$

$$\frac{\partial o_{11}}{\partial w^1_{11}} = x_1 \cdot w_{11} + x_2 \times w^1_{21} + b_{11} = x_1$$

$$\frac{\partial o_{11}}{\partial w^1_{11}} = x_1 \cdot w^1_{11} + x_2 \cdot w^1_{21} + b$$

$$= x_1$$

$$\# \frac{\partial L}{\partial w^1_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w^1_{12}}$$

$$\# \frac{\partial L}{\partial w^1_{22}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{12}} \frac{\partial o_{12}}{\partial w^1_{22}}$$

$$\# \frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial o_{12}} \frac{\partial o_{12}}{\partial b_{12}}$$

$$\frac{\partial \hat{y}}{\partial o_{12}} = w^2_{21}$$

$$\frac{\partial o_{12}}{\partial w^1_{12}} = \frac{\partial}{\partial w^1_{12}} \left(x_1 w^1_{12} + x_2 w^1_{22} \right)$$

$$= x_1$$

$$\frac{\partial o_{12}}{\partial w^1_{22}} = x_2$$

$$\frac{\partial o_{12}}{\partial b_{12}} = 1$$

Final:

$$\frac{\partial L}{\partial w^1_{11}} = -2(y-\hat{y})\, w^2_{11} \times x_1$$

$$\frac{\partial L}{\partial w^1_{21}} = -2(y-\hat{y})\, w^2_{11} * 2$$

$$\frac{\partial L}{\partial b_{11}} = -2(y-\hat{y})\, w^2_{11}$$

$$\frac{\partial L}{\partial w^1_{12}} = -2(y-\hat{y}) \cdot w^2_{21} \cdot x_1$$

$$\frac{\partial L}{} = -2(y-\hat{y})\, w^2_{21} \cdot x_2$$

∂w_{22}

$$\frac{\partial L}{\partial w_{21}^{2}} = -2(y-\hat{y})\, w_{21}^{2}$$

$$\frac{\partial L}{\partial w_{11}^{2}} = -2(y-\hat{y})\, o_{11}$$

$$\frac{\partial L}{\partial w_{21}^{2}} = -2(y-\hat{y})\, o_{12}$$

$$\frac{\partial L}{} = -2(y-\hat{y})$$